Quantifying environmental trends: Disentangling changes in impact from changes in monitoring

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Last Updated on January 09, 2025

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Table 1: Total global annual CO2 emissions (million metric tonnes) by fleet for 2016 and 2023. The percent of total 2023 emissions for each fleet is also shown, along with the percent change from 2016 to 2023 for each fleet.

fleet	2016	2023	percent_total_2023	percent_change
AIS-broadcasting	839	1110	84%	+33%
Non-broadcasting	250	209	16%	-17%
Total	1090	1320	100%	+21%

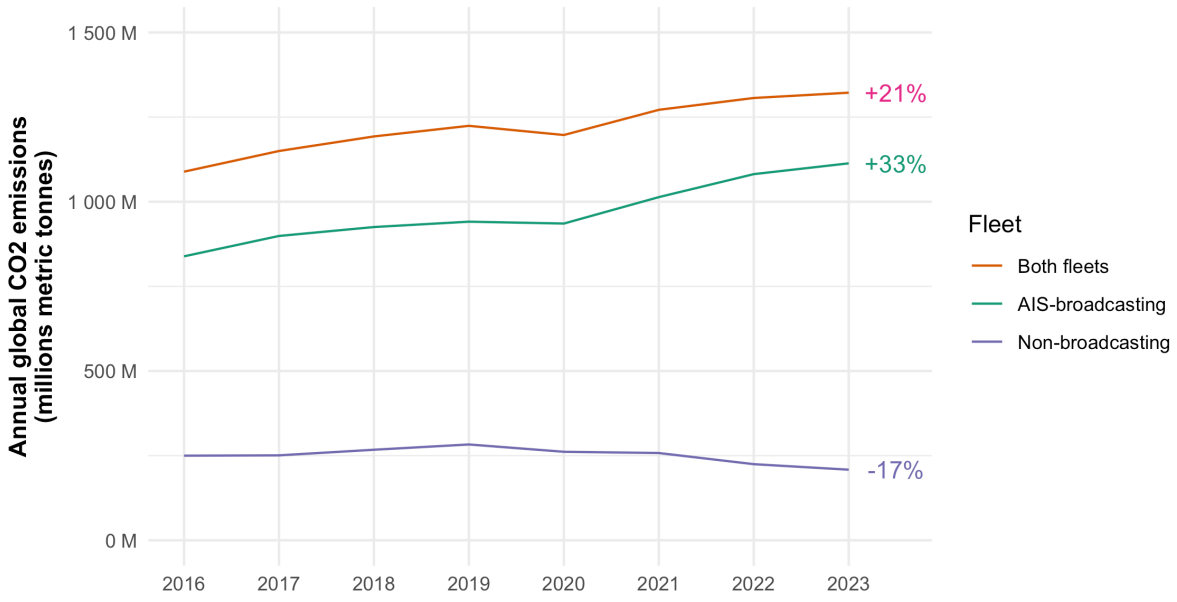

Figure 1: Total global annual CO2 emissions by fleet. The percent change from 2016 to 2023 for each fleet is shown next to its line.

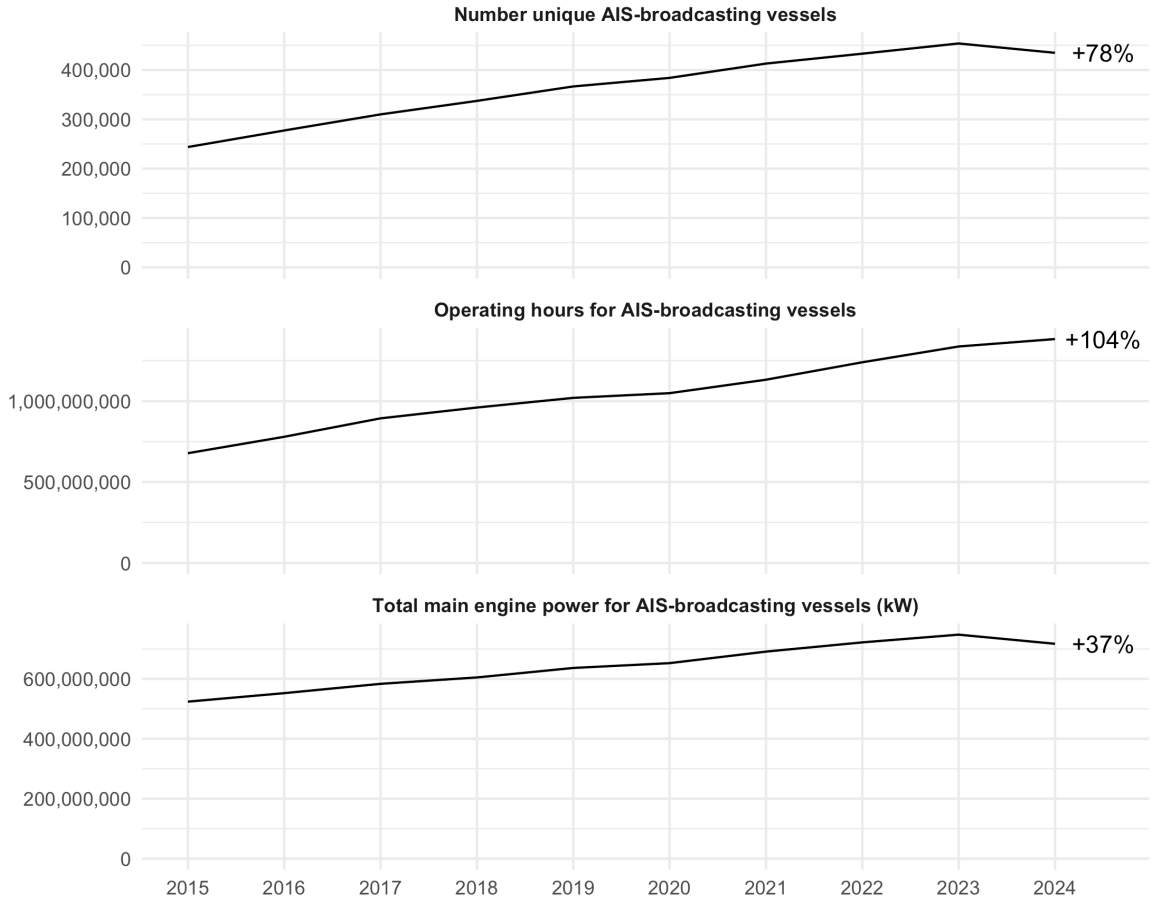

Figure 2: Annual unique number of AIS-broadcasting vessels (top panel); annual number of operating hours for AIS-broadcasting vessels (middle panel); and annual total engine power for those vessels (killowatt). The percent change from 2015 to 2024 is shown next to each line.

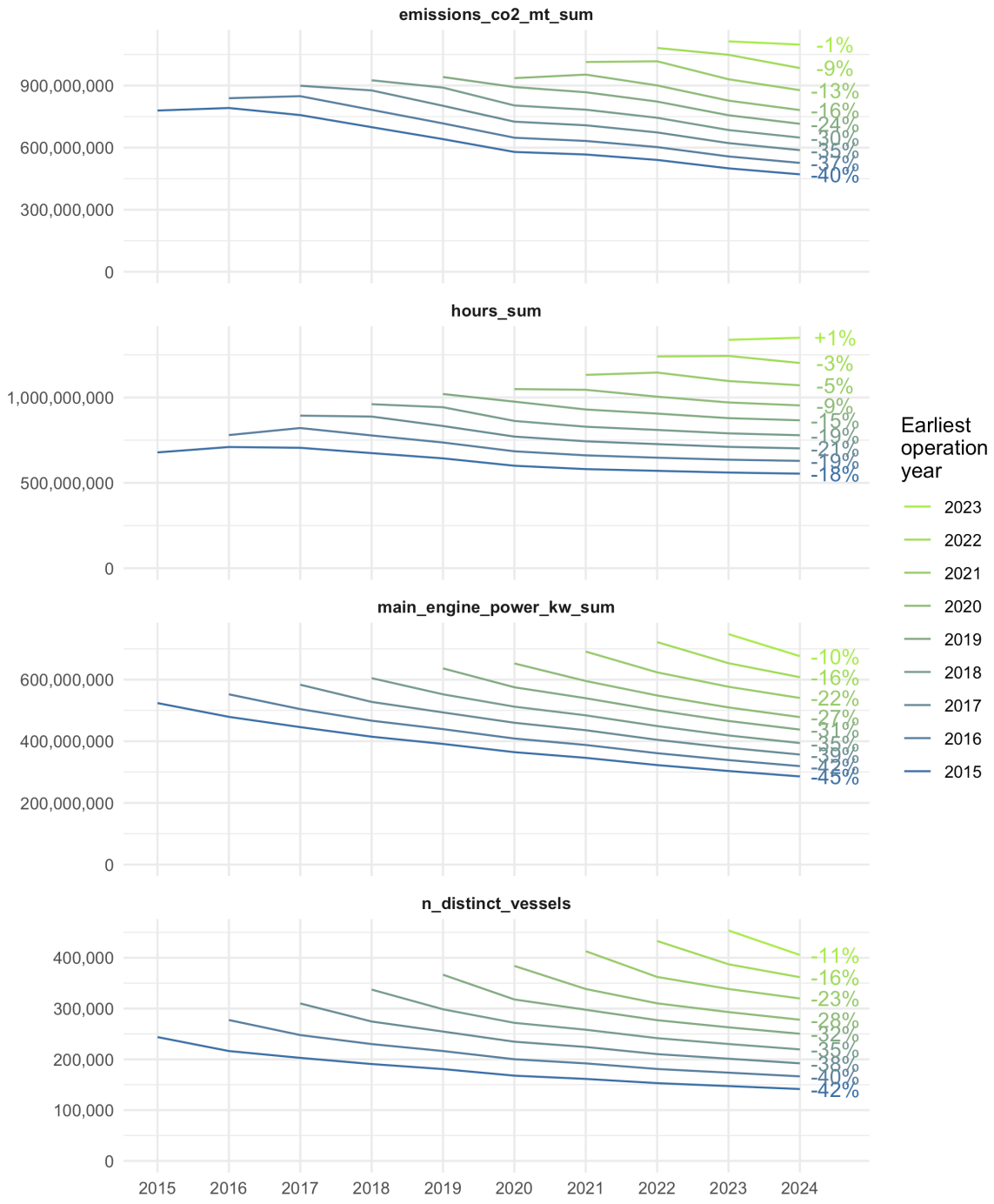

Figure 3: Total global CO2 emissions; number unique vessels; operating hours; and total main engine power for the set of AIS-broadcasting vessels that have been operating since a given year, where each line is colored by the earliest operation year. The percent change from the earliest year to 2023 is shown next to its line.

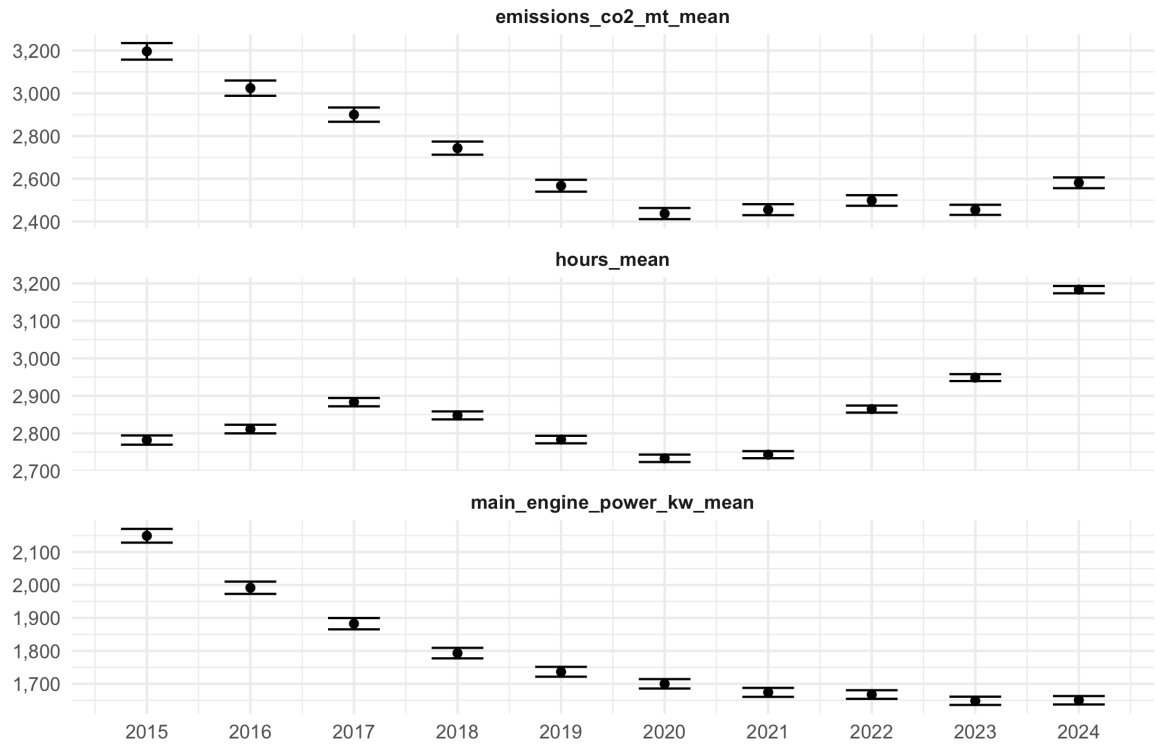

Figure 4: Annual distributions of individual per-vessel CO2 emissions (metric tonnes, top panel); operating hours (middle panel); and main engine power (panel). The point shows the mean value, and the error bars show +/- two standard errors.

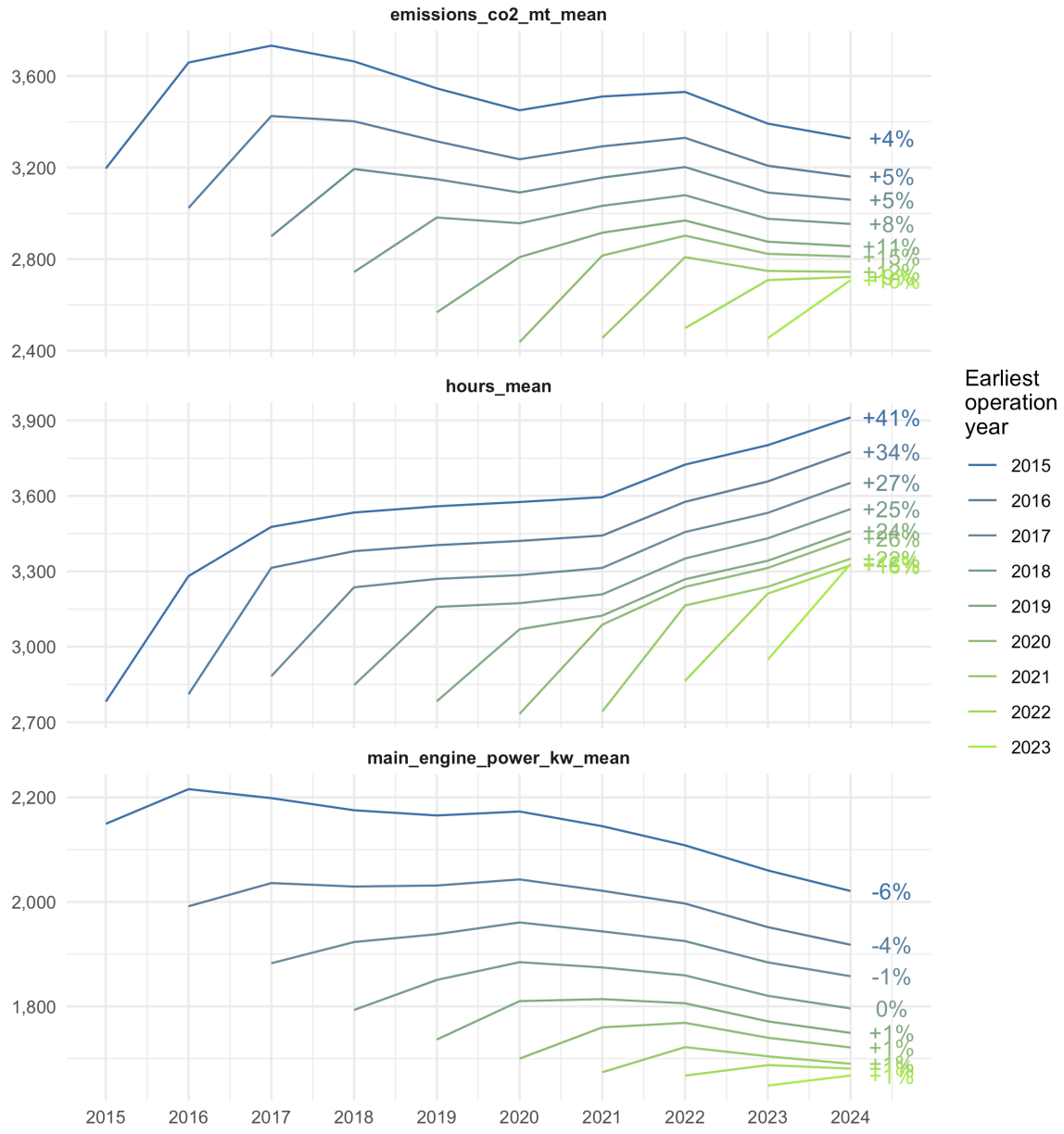

Figure 5: Annual mean of individual per-vessel CO2 emissions (metric tonnes, top panel); operating hours (middle panel); and main engine power (panel) for the set of AIS-broadcasting vessels that have been operating since a given year, where each line is colored by the earliest operation year. The percent change from the earliest year to 2023 is shown next to its line.

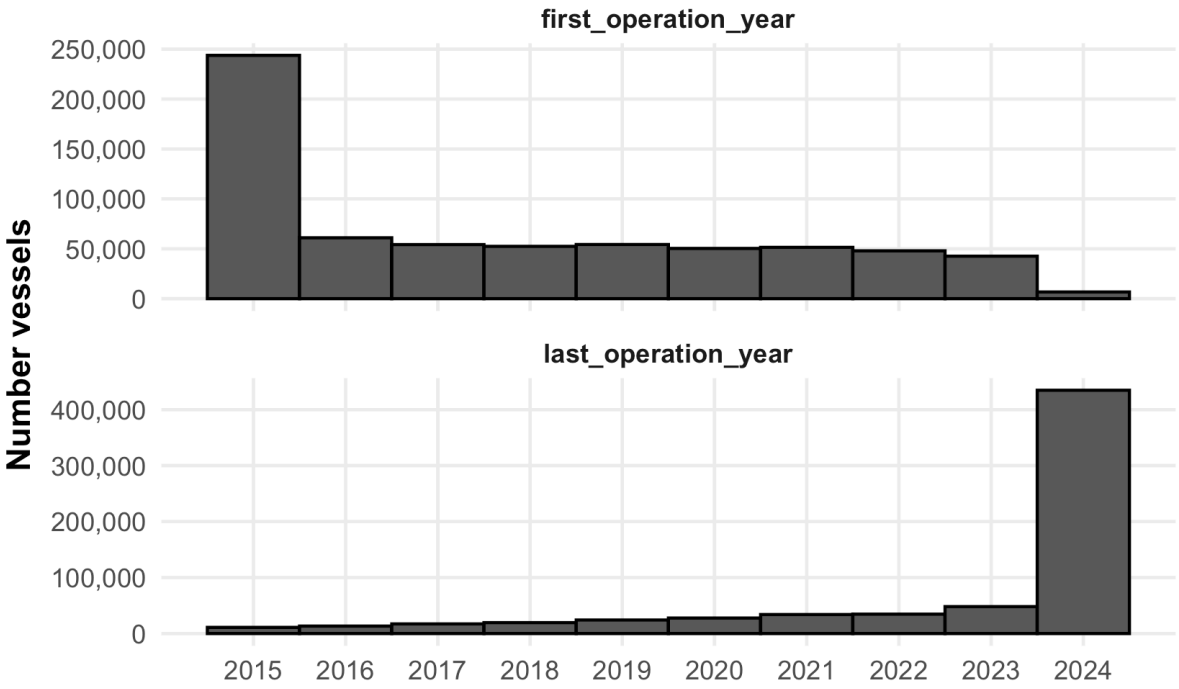

Figure 6: Histograms showing the distribution of the first and last operation year of each vessel.

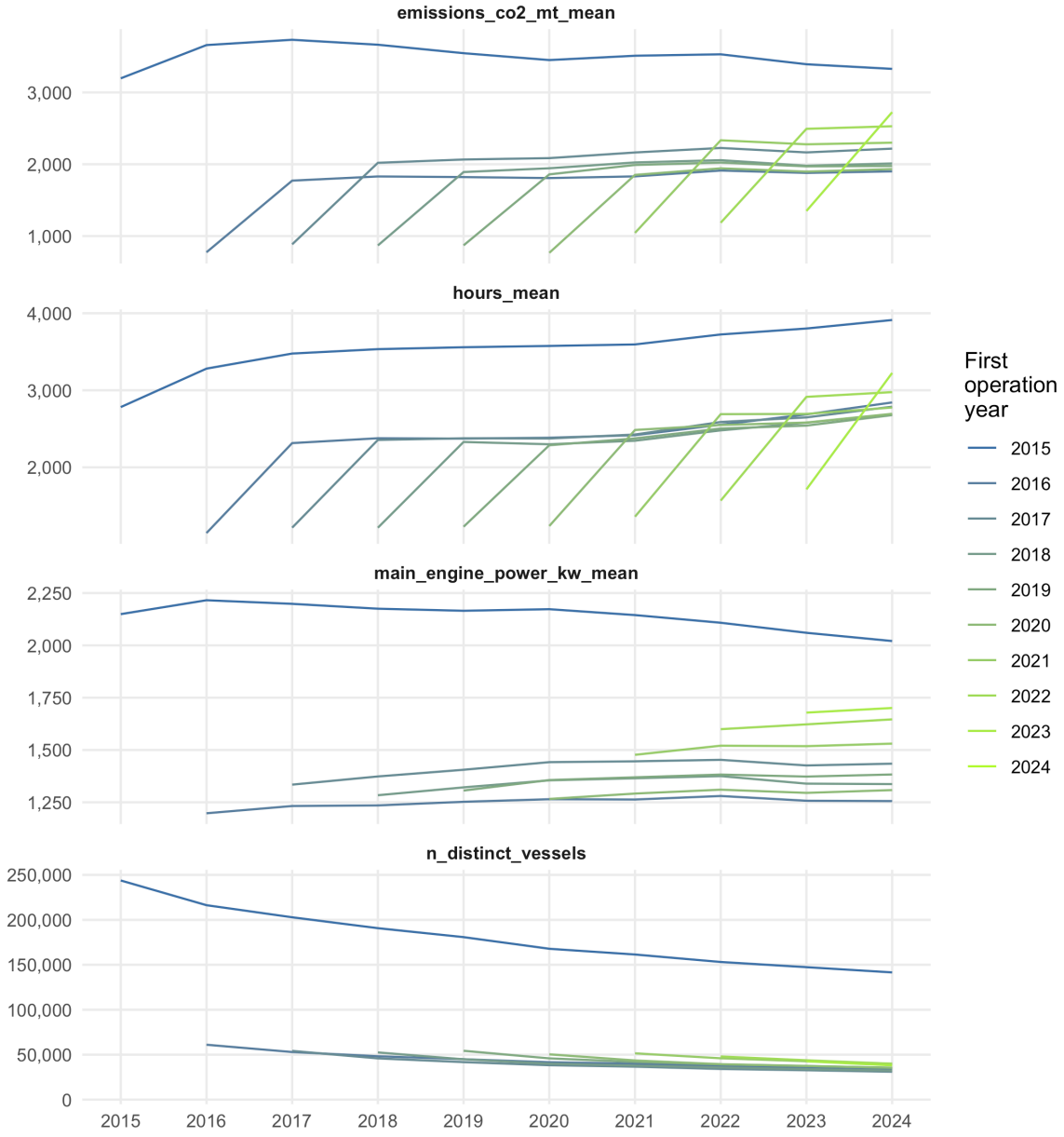

Figure 7: Annual mean of individual per-vessel CO2 emissions (metric tonnes); mean per-vessel operating hours; mean per-vessel main engine power; and number of distinct vessels for the set of AIS-broadcasting vessels for each first year of operation, where each line is colored by the first operation year.

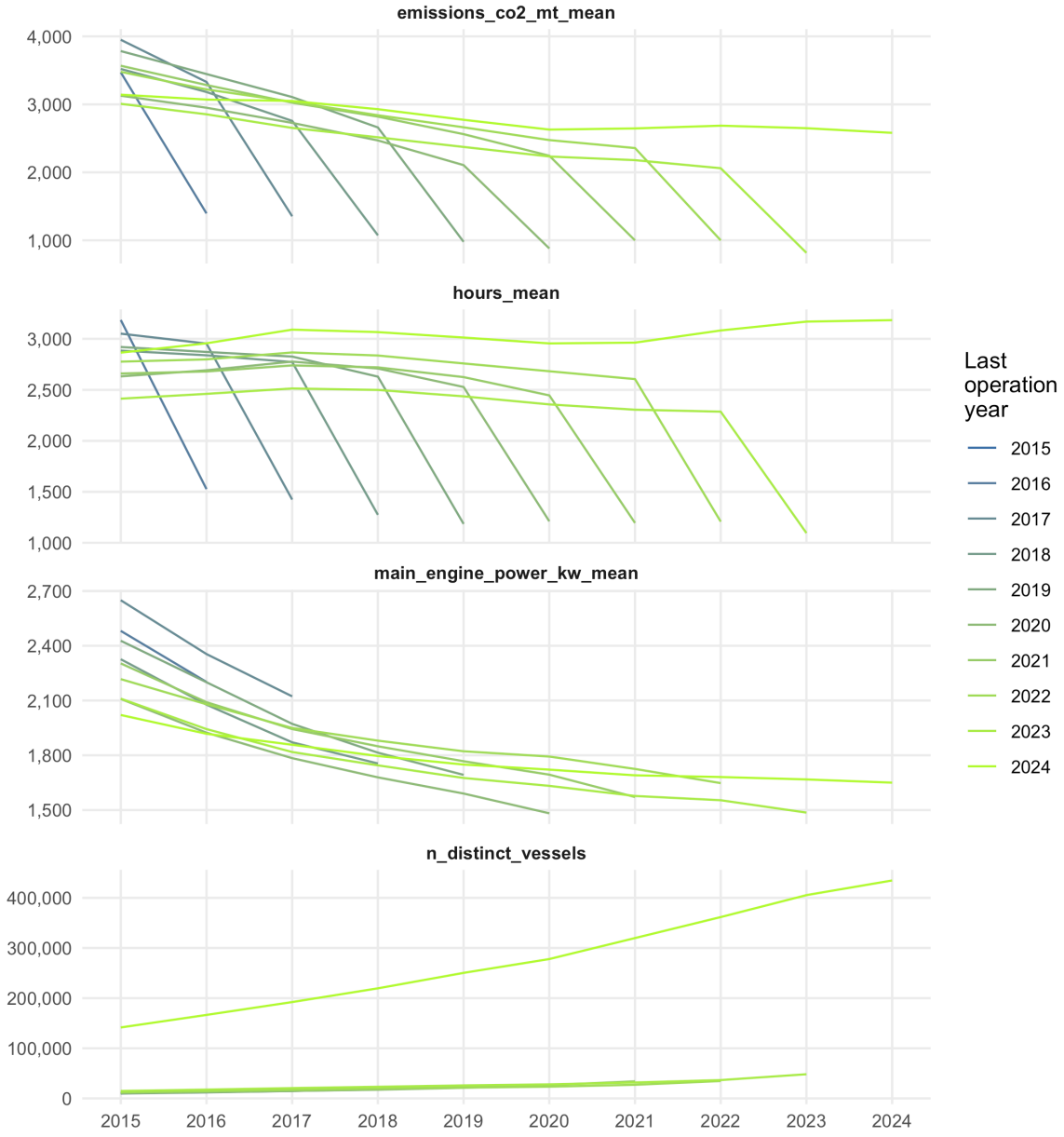

Figure 8: Annual mean of individual per-vessel CO2 emissions (metric tonnes); mean per-vessel operating hours; mean per-vessel main engine power; and number of distinct vessels for the set of AIS-broadcasting vessels for each last year of operation, where each line is colored by the last operation year.